

DC0001A	Computer Programming in C++	3:0:0	3
---------	-----------------------------	-------	---

OBJECTIVE:

- To perform object oriented programming solution and develop solutions to problems demonstrating usage of control structure, modularity, classes, I/O and the scope of the class members
- To demonstrate adeptness of object oriented programming in developing solution to problems demonstrating usage of data abstraction, encapsulation and inheritance
- To demonstrate ability to implement one or more patterns involving dynamic binding and utilization of polymorphism in the solution of problems
- To learn syntax and features of exception handling
- To demonstrate the ability to implement solution to various I/O manipulation operations and the ability to create two-dimensional graphic components using applets

UNIT 1	C++ Overview, C++ Characteristics, Object-Oriented Terminology, Polymorphism, encapsulation, inheritance, Object-Oriented Paradigm, Abstract Data Types, I/O Services, Standard Template Library, Standards Compliance, Functions and Variables. Declaration and Definition
UNIT 2	Variables: Dynamic Creation and Derived Data, Arrays and Strings in C++, Classes in C++, Defining Classes in C++, Classes and Encapsulation, Member Functions, Friend function, Inline function
UNIT 3	Using Constructors, Multiple Constructors and Initialization Lists, Using Destructors to Destroy Instances, Using Destructors to Destroy Instances, Operator Overloading: operator overloading of unary and binary operator, Function Overloading, Working with Overloaded Operator Methods
UNIT 4	Constant and Static Class Members, Inheritance, Overview of Inheritance,

	Defining Base and Derived Classes, Single, Multiple, multilevel, hybrid hierarchical inheritance. Constructor and Destructor Calls in inheritance, virtual function, virtual base class,
UNIT 5	Input and Output in C++ Programs, Standard Streams, Manipulators, Unformatted Input and Output. Working with files.

Course Outcome (CO):

At the end of this course, students will demonstrate ability to:

CO1: Understand object-oriented programming features in C++.

CO2: Apply these features to program design and implementation.

CO3: Develop applications using Object Oriented Programming Concepts.

CO4: Implement features of object oriented programming to solve real world problems.

CO5: CO5: Develop the ability to implement software in high-level programming language like C++

MAPPING COURSE OUTCOMES LEADING TO THE ACHIEVEMENT OF PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

Course Outcome	Program Outcome												Program Specific Outcome		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	M												M		
CO2			H		H									H	L
CO3			H		M				M		M		H		
CO4				L								M		L	
CO5	H		H		M			L	M				H	M	

Text Books

- Let Us C: BalaGuruswamy, TATA McGraw Hill.
- Programming with C, C++: Yashwant Kanetkar

Reference Books

- C++: The Complete Reference.
- The C++ Programming Language: Bjarne Stroustrup